TECHTO · FAST & FRUGAL · TEAM 15

One prompt in. Three hotels called. One email out.

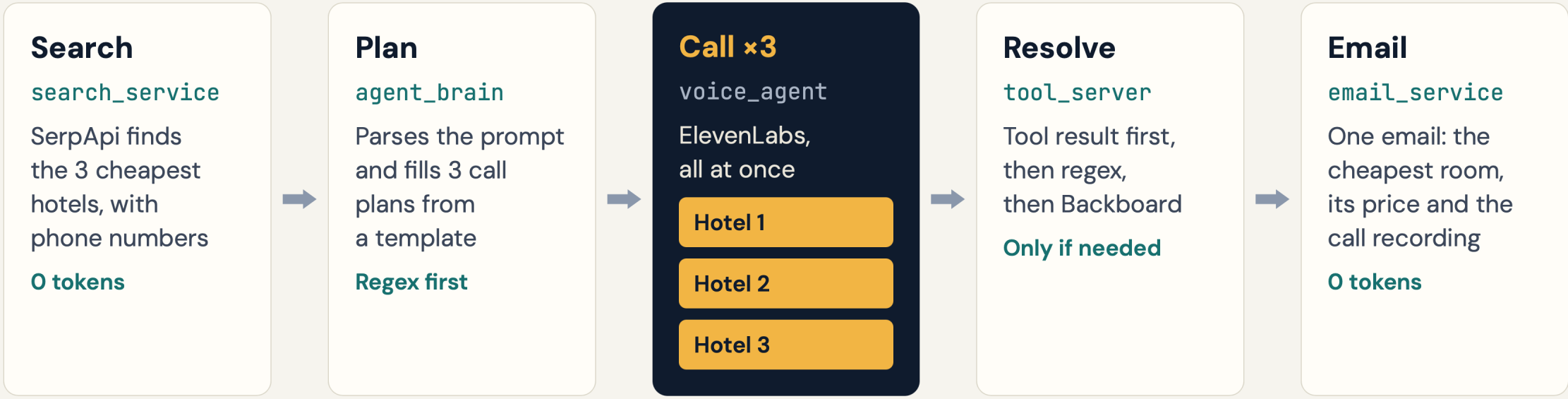
A voice agent phones the three cheapest hotels at the same time, asks whether a room is free, and emails the cheapest one with its price and a link to the call recording. It never books.



Isaac · WS4 · Confirmation extraction, email and pitch

ARCHITECTURE

Three calls run in parallel, and the email waits for the last one



Total time is the slowest call, not the sum of three. There is no failover: a voicemail is simply that hotel's result.

TOKEN ECONOMY

Tokens go only where language is needed

Stage	Tokens	Why
Search + phones	0	SerpApi, no LLM
Parse + plan	0, or ≤150	Regex and templates; Backboard if vague
Live calls ×3	~550 per call	Short prompt, one-sentence replies
Read the answer	0	Tool call, then regex
Fallback	≤300 per call	Only if the tool never fired
Email	0	Plain string formatting

< 2,700

Backboard tokens per session (budget)

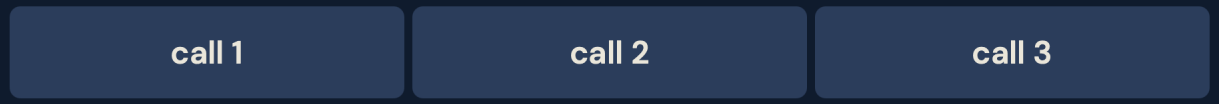
2,101

Agent tokens per 3-call run, measured (avg of 7 real runs)

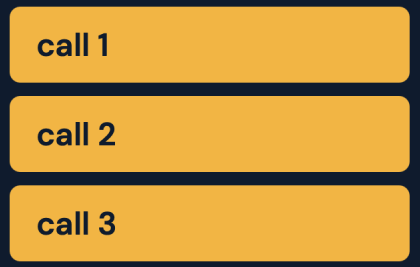
LATENCY POST-MORTEM

Parallel calls: total time is the slowest call

If we dialled one by one



We dial all three at once



Measured over 7 real runs: 70 s with all three at once, 130 s if dialled one by one.

Search	0.3s
Plan	1.4s
Voicemail caught	14 s avg
Calls, 3 at once	70 s avg
Email send	0.2 s
Total (target < 4 min)	**87.9s**

CALL HANDLING

Menus, holds and pushy desks, without booking

Menus

Says “reservations” or presses 1, and tries 0 for the front desk.

Holds

Stays silent through hold music. A long silence timeout, not a slow turn pause.

Voicemail

Spotted from the greeting, ended within 45 s, and reported as Unknown.

The answer

Repeated back, then a structured tool call. Regex and Backboard are backups.

Never books

Declines names and cards: “I’m only checking availability today.”

Reservation

If a hotel volunteers a reservation number, it goes in the email. We never ask for one.

Tested on 52 real calls to our own phones, in 21 runs

LEARNINGS

What we changed, and what's next

Backboard before and after the calls, not during
A model hop on every turn adds delay to every reply.

A structured tool call, not transcript regex
Speech-to-text mangles spoken numbers and prices.

Availability only: the agent never books
We must not create real reservations at real hotels.

Dial all three at once, with no failover
Total time is the slowest call, not the sum.

One email, after the last call
It needs all three results before it can pick the cheapest room.

Next

- Deploy the tool server so we don't depend on a tunnel
- Keep results in storage, so a restart can't lose them